

AUTONOMOUS DATA SCIENTIST

AUTODS FINAL DATA SCIENCE REPORT

DATASET UPLOAD

The uploaded dataset was accepted and passed to the AutoDS pipeline.

1. EXECUTIVE SUMMARY

The AutoDS pipeline profiled, cleaned, explored, modeled and visualized the uploaded dataset, then consolidated the findings into this report.

2. PROBLEM STATEMENT

The analysis evaluates dataset quality, relationships between variables, predictive performance and actionable findings using the uploaded data.

3. DATASET OVERVIEW

Rows: 119

Columns: 2

Columns: ['Experience', 'Salary']

4. DATA QUALITY ASSESSMENT

Missing Values: 0

Duplicate Rows: 0

5. DATA CLEANING

Original Rows: 120

Cleaned Rows: 119

Duplicates Removed: 1

6. EXPLORATORY DATA ANALYSIS

Numerical Columns:

['Experience', 'Salary']

Categorical Columns:

[]

Statistics:

Experience: mean=5.789915966386555, median=5.8, minimum=1.1, maximum=10.0, standard deviation=2.5406323434812275

Salary: mean=9.124033613445379, median=8.25, minimum=2.22, maximum=19.87, standard deviation=5.169212856729988

Correlation:

Experience: Experience=1.0, Salary=0.9751

Salary: Experience=0.9751, Salary=1.0

8. BEST MODEL

Algorithm:

Linear Regression

Feature:

Experience

Target:

Salary

Training Rows:

95

Testing Rows:

24

MAE:

0.9914939878132228

RMSE:

1.1807813682562762

R2 Score:

0.9521878040458659

9. MODEL EVALUATION

MAE: 0.9914939878132228

RMSE: 1.1807813682562762

R2 Score: 0.9521878040458659

7. VISUALIZATION INSIGHTS

Library: Plotly

Plots Generated: 6

10. KEY DATA INSIGHTS

- The dataset contains no missing values.
- No duplicate rows were found.
- Correlation between Experience and Salary is 0.9751.
- There is a strong positive relationship between the first two numerical variables.
- A Linear Regression model was trained using the first numerical column to predict the second.
- Plotly visualizations were generated for distributions, relationships, correlations and outliers.

11. BUSINESS RECOMMENDATIONS

- Validate the model using unseen data.
- Perform cross-validation.
- Compare multiple ML algorithms.
- Add useful features.
- Monitor model performance.
- Analyze generated visualizations.

12. LIMITATIONS

The modeling result uses the available numerical columns and a single train/test split. Results should be interpreted within the scope and quality of the uploaded dataset.

13. FUTURE IMPROVEMENTS

- Compare additional model families.
- Use cross-validation and broader feature engineering.
- Validate recommendations with domain knowledge and new data.

14. CONCLUSION

The AutoDS system successfully processed the dataset through profiling, cleaning, EDA, modeling, visualization, AI insights and report generation.

END OF REPORT